

Muzammil Nawaz Khan

AI Engineer

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SUMMARY

AI Engineer with 1+ year of production experience building machine learning, LLM and RAG systems in Python. Delivered a claims-classification system at 95% accuracy across 42,900+ cases, cut a claim denial rate from 3.2% to 2.4%, and cut inference costs 35% through prompt caching. Computer vision background in 3D MRI brain tumor segmentation (SegFormer3D, 85%+ Dice). Skilled in PyTorch, TensorFlow, scikit-learn, LangChain, FastAPI and Docker. Open to remote work and relocation.

SKILLS

Generative AI & Agents: OpenAI API, Anthropic API, LLMs, prompt engineering, structured outputs, RAG, LLM-as-judge evaluation, LangChain, LangGraph, CrewAI, AutoGen, Model Context Protocol (MCP), n8n, ChromaDB, FAISS, custom HNSW, BM25, Reciprocal Rank Fusion

Machine Learning: Python, PyTorch, TensorFlow, Keras, scikit-learn, CatBoost, XGBoost, AutoGluon, Hugging Face Transformers, Pandas, NumPy

Voice & Real-Time: LiveKit, WebRTC, ElevenLabs, Silero VAD, STT/TTS, Twilio, SIP trunking, WebSockets, SSE

Computer Vision: OpenCV, U-Net / 3D U-Net, SegFormer / SegFormer3D, YOLO, GradCAM, medical image segmentation

Backend & Infrastructure: FastAPI, Django / DRF, REST APIs, Docker, Kubernetes, Git/GitHub, CI/CD

Domain: Healthcare IT, medical imaging, AI training data

EXPERIENCE

Data Trainer, SME Solutions, Inc. (via Deel)

Aug 2026 – Present

Remote, independent contractor

- Create and review detailed prompts and responses used to train AI models across varied topics.
- Evaluate and rank AI model responses to improve accuracy in different contexts.
- Test models for inaccuracies and bias, validating their applicability in the target domains.

Junior AI Engineer, CareCloud (MTBC)

Jul 2025 – Aug 2026

Islamabad, Pakistan

- Delivered a production claims-classification system (CatBoost, XGBoost, AutoGluon) at 95% parent-model accuracy and 92%+ category-level accuracy across 42,900+ cases, with daily feedback-loop retraining and multi-month tracking of accuracy and stability.
- Built a Python denial-prediction and resubmission automation system with a rule-based First-Time Pass Rate (FTPR) module comparing rejected vs corrected/paid claims by provider, payer, CPT and ICD code. Cut resubmission wait from 2 weeks to 3 days and reduced the denial rate from 3.2% to 2.4%.
- Designed an n8n workflow acting as a Model Context Protocol (MCP) server for conversation state, backend prompts and call routing, wrapping APIs as tool calls. Cut inference costs 35% through prompt caching.
- Worked across a real-time voice AI front-desk agent on LiveKit (WebRTC) with ElevenLabs TTS, OpenAI models and Silero VAD for low-latency interruption handling; Twilio/SIP trunking for inbound calls with transfer to human staff; LangGraph for multi-state orchestration. Reduced front-desk handling time by 40%.
- Layered rule-based checks with an LLM validation agent that generates root-cause analysis and corrective-action recommendations for each flagged case.
- Engineered Python pipelines that pull Hikvision NVR logs, clean them and load them to a database for HR attendance analytics; built an internal FastAPI EDI Viewer for inspecting raw healthcare claim data.

Software / AI Engineer, Suidhaga

Mar 2022 – Nov 2025

Remote, part-time alongside studies

- Progressed through the full engineering stack over 3+ years for an e-commerce business: built frontend UI, moved to Django-focused backend development, then owned full-stack web features end to end.
- Applied data analysis and data science to business data; later built ARIMA-family forecasting models to predict product prices.

Freelance AI Engineer, Independent
Remote

Sep 2022 – Present

- Delivered AI development projects for clients, including model training and applied AI application builds.

AI/ML Research Intern, Risetech
Rawalpindi / Islamabad, Pakistan

Jul 2024 – Aug 2024

- Benchmarked 3D U-Net against SegFormer3D and selected SegFormer3D (85%+ Dice). Used GradCAM for explainability and FastAPI for serving.
- The work became the Neurofusion final year project: 1st Place at COMPPEC 2025 and a Departmental Silver Medal.

PROJECTS

Verifiable Agent Kernel (VAK) (Rust, Python, WASM, Cedar/ABAC): Deterministic control plane for LLM agents: least-privilege ABAC, WASM-sandboxed tools, Z3 formal verification before execution, 3-tier memory (Merkle DAG), multi-agent debate with quadratic-voting consensus, and a PRM fine-tuning toolkit. v1.0 with 80%+ test coverage; CI/CD on Docker and Kubernetes; LangChain/AutoGPT/MCP adapters. [GitHub: github.com/muzammil5539/Verifiable-Agent-Kernel-VAK](https://github.com/muzammil5539/Verifiable-Agent-Kernel-VAK)

RAG Custom Engine (Pure Python, GPT-4o-mini): From-scratch hybrid retrieval with a custom HNSW store, Okapi BM25 and Reciprocal Rank Fusion, plus a 3-stage Self-RAG gate (retrieval decision, relevance grading, hallucination check) and contextual compression. About 20% precision gain over dense-vector baselines.

Conversational AI Agent (ReAct) (LangGraph, FastAPI, Next.js, OpenAI): Six tools (Calculator, Weather, DateTime, Web Search, Code Interpreter, RAG Search), visible reasoning traces, WebSocket streaming and a SQLite checkpoint for memory.

Document Summarizer (FastAPI, Next.js 14, LangChain, ChromaDB): Automatic strategy router (Stuff, Map-Reduce, Refine, Recursive), SSE token streaming and async pipelines. About 50% lower latency and 99% factual consistency.

RAG LangChain Chroma (LangChain, ChromaDB, FastAPI): Hybrid RAG application with dual-layer memory: in-chat session memory and cross-chat LLM-summarized memory.

EDUCATION

Bachelor of Computer Engineering, National University of Sciences and Technology (NUST)
Islamabad, Pakistan

Nov 2021 – Jul 2025

CGPA 3.25. Final year project: Neurofusion, 3D brain tumor segmentation (SegFormer3D, BraTS20, 85%+ Dice); 1st Place COMPPEC 2025, Departmental Silver Medal.

F.Sc. Pre-Engineering, Ghazali Premier College
Lahore, Pakistan

Jun 2018 – Aug 2020

CERTIFICATIONS

- IBM AI Engineering Professional Certificate
- IBM Generative AI Engineering Professional Certificate
- IBM RAG and Agentic AI Professional Certificate
- IBM Deep Learning with PyTorch, Keras and TensorFlow Professional Certificate
- Generative AI Engineering with LLMs Specialization (IBM)
- Large Language Model Operations (LLMOps) Specialization, Duke University
- Building GenAI Applications and Agents Specialization, Coursera
- RAG for Generative AI Applications

HONORS & AWARDS

1st Place, COMPPEC 2025 | Departmental Silver Medal